

InfraRisk AI - Comprehensive Project Report

Repository: vikashg450/Data-Scientist-InfraRisk-AI

Status: Complete & Production-Ready

Date: June 12, 2026

Executive Summary

InfraRisk AI is a sophisticated, production-grade multi-modal AI platform designed to quantify credit risk and support credit decisioning for cross-border infrastructure project finance. The system combines:

- â€¢ Geospatial computer vision - Sentinel-2 satellite imagery analysis
- â€¢ Macroeconomic modeling - World Bank data, commodities, FX rates
- â€¢ Deep learning ensembles - CNN, TFT, GNN, PINN, Legal-BERT
- â€¢ Financial analytics - DSCR, leverage, cash flow projections
- â€¢ Interactive simulation cockpit - Gamified risk management

Key Statistics

- â€¢ Total Code: ~372,670 lines (86.5% Python, 13.9% Jupyter, 0.3% Docker)
- â€¢ Completion Rate: 100% (all 8 components delivered)
- â€¢ Deployment: Docker containerized (Streamlit + MLflow + Feast)

Project Objectives

1. Quantify Infrastructure Credit Risk - Predict PD and EL for infrastructure assets
2. Multi-Modal Feature Fusion - Integrate satellite, macro, financial, legal data
3. Real-Time Credit Decisioning - Support committees with transparent risk assessments
4. Portfolio Stress Testing - Model contagion across dependent assets
5. Interactive Risk Simulation - Gamified cockpit for risk managers
6. Production MLOps - MLflow, DVC, Great Expectations, GitHub Actions

Core AI Models

1. CNN Progress Monitor - Siamese ResNet-50 (MAPE < 15%)
2. Demand Forecaster - Temporal Fusion Transformer with quantile regression
3. Portfolio GNN - PyTorch Geometric contagion model
4. PINN Degradation - Physics-informed neural network for asset lifecycle
5. Legal-BERT - Contract parsing and clause classification
6. XGBoost/LightGBM - Gradient boosting credit scorers (100+ features)

Final Outputs

- â€¢ PD (Probability of Default): 0-100%, calibrated to historical rates
- â€¢ EL (Expected Loss): % of asset value at risk
- â€¢ SHAP Explainability: Feature importance rankings, force plots
- â€¢ Gamified Dashboard: Streamlit interactive cockpit

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Status: Production-Ready | All Components Delivered | 100% Complete